

Affine Transforms, Three Covariances, and Borel Clusters of an Exponential Hawkes Process

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Abstract

For the exponential Hawkes process, we close the joint count–intensity transform, the stationary intensity law, and a triangular recurrence for all moments. The subcritical condition is treated together with its parameter faces. We then separate intensity covariance, complete counting covariance (including its Dirac atom), and the Bartlett spectrum under a frozen Fourier convention, obtaining exact window-count variance. All results are source-local; no arithmetic or target spectral data enter.

1 The predictable Markov owner

Let $b > 0$, $a, \nu \geq 0$, and let N be a simple point process whose predictable event rate is the left limit λ_{t-} , where

$$d\lambda_t = -b(\lambda_t - \nu) dt + a dN_t. \quad (1)$$

Thus an event is sampled with the pre-jump intensity, then raises the post-jump intensity by a . Its Markov generator is

$$\mathcal{G}f(x) = b(\nu - x)f'(x) + x\{f(x + a) - f(x)\}. \quad (2)$$

This convention prevents the common but consequential confusion between λ_{t-} and λ_t at an event.

Theorem 1 (joint affine transform). *For $0 \leq z \leq 1$, $s \geq 0$, and $\lambda_0 = x$,*

$$\mathbb{E}_x[z^{N_t - N_0} e^{-s\lambda_t}] = e^{-A_t - B_t x}, \quad (3)$$

where $B_0 = s$, $A_0 = 0$, and

$$B'_t = 1 - bB_t - ze^{-aB_t}, \quad A'_t = b\nu B_t. \quad (4)$$

Proof. In the backward equation, a jump multiplies the transform by z . Substituting $F(t, x) = e^{-A_t - B_t x}$ into (2), with $f(x + a)$ replaced by $zf(x + a)$, gives constant coefficient $-b\nu B_t$ and x -coefficient $bB_t + ze^{-aB_t} - 1$. Equating these with $-A'_t - B'_t x$ proves (4). The transform is bounded on the stated parameter domain, so localization and bounded convergence close the calculation. $\square$

2 Stationarity, Laplace law, and every moment

Suppose first that $a < b$, set $\delta = b - a$, and put $\mu = b\nu/\delta$. The excitation kernel is $h(t) = ae^{-bt}\mathbf{1}_{t>0}$ and has mass $a/b < 1$.

Theorem 2 (stationary closure). *For $\nu > 0$ there is a unique finite-intensity stationary Hawkes law. Its intensity Laplace transform $L(s) = \mathbb{E}e^{-s\lambda}$ is the unique normalized solution*

$$\frac{L'(s)}{L(s)} = -\frac{b\nu s}{bs + e^{-as} - 1}, \quad L(0) = 1. \quad (5)$$

Writing $m_n = \mathbb{E}\lambda^n$ and $m_0 = 1$, every moment is finite and

$$m_n = \frac{nb\nu m_{n-1} + \sum_{k=0}^{n-2} \binom{n}{k} a^{n-k} m_{k+1}}{n\delta}, \quad n \geq 1. \quad (6)$$

In particular, $\mathbb{E}\lambda = \mu$ and $\text{Var } \lambda = \mu a^2/(2\delta)$.

Proof. The immigration–Poisson-cluster construction is subcritical, hence locally finite and stationary; the same genealogy gives uniqueness among finite-intensity stationary Hawkes processes. Applying stationary (2) to e^{-sx} gives

$$0 = -b\nu sL(s) - \{bs + e^{-as} - 1\}L'(s).$$

The bracket has derivative $b - ae^{-as} > 0$ and vanishes only at zero, so (5) has one normalized solution. Applying (2) to x^n , expanding $x\{(x+a)^n - x^n\}$, and isolating the coefficient $n(b-a)m_n$ gives (6). Induction closes every moment, and its first two instances give the displayed mean and variance. $\square$

3 Three covariances and exact window variance

The following objects have different dimensions and must not be conflated. The first is a covariance function of the stochastic intensity; the second is a measure for point counts and contains a same-event atom; the third is the Fourier transform of that complete measure.

Theorem 3 (covariance separation). *In the subcritical stationary regime,*

$$C_\lambda(t) = \text{Cov}(\lambda_t, \lambda_0) = \frac{\mu a^2}{2\delta} e^{-\delta|t|}, \quad (7)$$

$$\Gamma(dt) = \mu\delta_0(dt) + \frac{\mu a(2b-a)}{2\delta} e^{-\delta|t|} dt. \quad (8)$$

Under the convention with no $1/(2\pi)$ factor,

$$S(\omega) := \int_{\mathbb{R}} e^{-i\omega t} \Gamma(dt) = \mu \frac{b^2 + \omega^2}{\delta^2 + \omega^2}. \quad (9)$$

Consequently, for a stationary count in a window of length T ,

$$\text{Var}(N_T - N_0) = \mu T + \mu a(2b-a) \left\{ \frac{T}{\delta^2} - \frac{1 - e^{-\delta T}}{\delta^3} \right\}. \quad (10)$$

Proof. Conditional first moments solve $u'(t) = b\nu - \delta u(t)$, so $\mathbb{E}(\lambda_t \mid \lambda_0 = x) = \mu + (x - \mu)e^{-\delta t}$. Multiplication by $\lambda_0 - \mu$ and the variance above give (7).

For counts, Hawkes’s complete point spectrum is $\mu/|1 - a/(b + i\omega)|^2$ [1]. Algebra gives

$$S(\omega) = \mu + \frac{\mu a(2b-a)}{\delta^2 + \omega^2}.$$

Because $e^{-\delta|t|}$ transforms to $2\delta/(\delta^2 + \omega^2)$ in the frozen convention, inversion gives (8), including the indispensable $\mu\delta_0$ term. Finally integrate Γ over $[0, T]^2$: the atom contributes μT , while the continuous part is twice its positive-lag coefficient times $\int_0^T (T-t)e^{-\delta t} dt$. This is (10), whose long-window slope is $S(0) = \mu b^2/\delta^2$. $\square$

object	includes same-event atom?	coefficient of $e^{-\delta t }$
intensity covariance $C_\lambda(t)$	not applicable	$\mu a^2/(2\delta)$
counting covariance Γ	yes, $\mu\delta_0$	$\mu a(2b-a)/(2\delta)$
Bartlett spectrum $S(\omega)$	atom becomes μ	Fourier-domain rational law

References

- [1] A. G. Hawkes, *Spectra of some self-exciting and mutually exciting point processes*, Biometrika **58** (1971), 83–90, [doi:10.1093/biomet/58.1.83](https://doi.org/10.1093/biomet/58.1.83).